**LOK JAGRUTI UNIVERSITY (LJU)**
**INSTITUTE OF ENGINEERING & TECHNOLOGY**

**Department of Computer Engineering (701)**
**Bachelor of Technology (B.E.) – Semester - III**

| | | Teaching Scheme | | | | |
|---|---|---|---|---|---|---|
| **Course Code:** | **117013392** | | | | | |
| **Course Name:** | **Full Stack Development with Javascript-1** | **Lecture (L)** | **Tutorial (T)** | **Practical (P)** | **Credit** | **Total Hours** |
| **Category of Course:** | Engineering Science Course (ESC) | **3** | **0** | **6** | **6** | **30** |

**Syllabus**

| Unit No. | Topic | Prerequisite Topic | Successive Topic | Teaching Hours |
|---|---|---|---|---|
| 01 | **Introduction to Web and Web design** | | | 1 (5%) |
| | 1.1 Basics of WWW, HTTP Protocol: Features, Request and Response, Various status codes, URL, Client Server architecture | --- | --- | |
| | 1.2 Concepts of Effective Web Design:Website Planning, **Web design issues:** Browser Compatibility, Bandwidth and Cache, Display Resolution, Look and Feel of Website, Page Layout and Linking, Designing Effective Navigation | --- | --- | |
| 02 | **HTML Structure and Core Tags** | | | 3 (10%) |
| | 2.1 **Basics of HTML:**HTML Page Structure, Building Blocks of HTML:<html>,<head>,<title>,<body>, Editors(Notepad, Notepad++, Sublime Text, VS Code) | --- | --- | |
| | 2.2 Semantic and Non-Semantic HTML Tags, Block and Inline Elements, Comments in HTML<br>**Header Tags:**<link>-favicon, <script>, <style><br>**<meta>:** name, content, charset, http-equiv, viewport, description, keywords, author<br>**Layout & Sectioning Tags:** ,  ,<header>, <footer>, <main>, <section>, <article>, <aside>, <nav><br>**Content & Text Tags:** <h1> to <h6>, <p>, <pre>,<br>, <hr><br>**Hyperlink Tag:** <a><br>**Text Formatting Tags:** <b> ,, <i> ,, <u> ,<mark>,<del>,<ins>, <sub>, <sup>, <abbr> | --- | --- | |
| | 2.3 **Media Tags:**<br><img>, <figure> ,<figcaption> ,<audio>, <video> ,<source><br>**<iframe>:** map, youtube, any web resources<br>**List tags:** <ul> ,<ol>, <li> **Attributes**: type, start, reversed<br>**Table tags:** <table>, <tr>, <td> ,<th> ,<thead>, <tbody>, <tfoot>, <caption> **Attributes**: colspan, rowspan | --- | --- | |
| 03 | **Forms, Graphics, and Interactive Elements in HTML** | | | 3 (10%) |
| | 3.1 **Form & Input Tags:**<form>, <fieldset>, <legend>,<input>, <textarea>, <button>, <label>, <datalist>, <select> ,<option> ,<optgroup><br>**<input> type:** text, password, email, number, **tel**, checkbox, radio, submit, reset, button, file, **hidden, date, range, color**<br>**Attributes:** action, method, type, name, id, value, placeholder, required, readonly, disabled, checked, maxlength, minlength, step, min, max, multiple, accept, autocomplete, autofocus, rows, cols, selected, list | --- | --- | |
| | 3.2 **SVG:** Line, Circle, Rectangle , Logo, Polygon(Star) | --- | --- | |
| | 3.3 **HTML Image Maps:** <map> and <area> | --- | --- | |
| 04 | **Introduction to CSS** | | | 3 (10%) |
| | 4.1 Significance of CSS,Types of CSS-Inline, Internal, External | --- | --- | |
| | 4.2 **Various CSS Selectors:** Id, Class, Attribute, Universal, Descendant, Child , grouping selector, Pseudo class, Pseudo element<br>CSS Specificity explanation | --- | --- | |
| | 4.3 **Background Properties:** background-color, background-image, background-repeat, background-position, background-size, background-attachment, background-clip, background shorthand<br>**Text Properties:** color, text-align, text-indent, text-transform, text-decoration, letter-spacing, word-spacing, line-height, text-shadow<br>**Font Properties:** font-family, font-size, font-style, font-weight, font-variant, font , Google Fonts (via <link> or @import) | --- | --- | |
| | 4.4 **Border**: border-style, border-width, border-color, border<br>**Boxes**: margin, padding, width, height, box-sizing, box-shadow<br>**Display:** inline, block, inline-block, none, flex, grid<br>**Flexbox :** flex-direction, justify-content, align-items, flex-wrap, gap, order, flex (shorthand)<br>**Grid Layout** : grid-template-columns, grid-template-rows, gap (grid-gap), grid-column, grid-row<br>**Positioning**: position (static, relative, absolute, fixed), top, right, bottom, left, z-index | --- | --- | |
| 05 | **CSS Effects, Animations & Responsive Design** | | | 4 (10%) |
| | 5.1 **Multiple column:** column-count, column-gap, column-rule-style,<br>**Style images:** Rounded Images, Thumbnail, Responsive Images, Transparent Images<br>**Gradients:** Linear, Radial, Conic<br>**Transformations:** transform (rotate(), scale(), scaleX(), scaleY(), translate(), translateX(), translateY(), skew(), skewX(), skewY())<br>**Transitions:** transition-property, transition-duration, **transition-timing-function**, transition-delay<br>**Animations:** @keyframes, animation-name, animation-duration, animation-timing-function, Iteration Count, Direction | --- | --- | |
| | 5.2 **CSS Variables: :**root, var(),**Media Queries:** Syntax, Media Types | --- | --- | |

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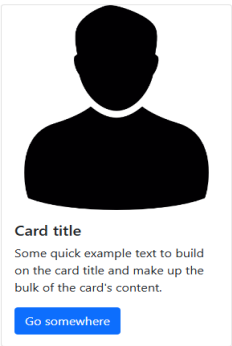
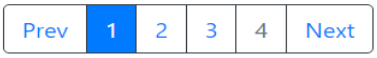
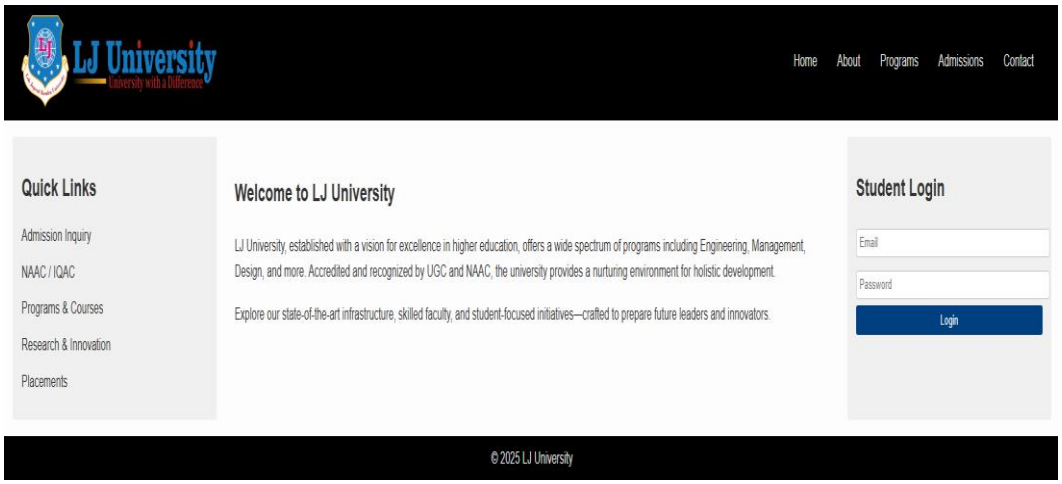
|---|---|---|---|---|
| 06 | **Introduction to Bootstrap & Utility Classes** | | | 3 (10%) |
| | 6.1 Bootstrap Setup, **Containers:** Fixed, Fluid,<br>**Utility first classes:**Spacing, Display, Colors, Sizing, Borders, Positioning, Typography, Shadows, Background, Text, Align, Margin, Padding,Flex<br>**Images**: Rounded, Circle, Thumbnail, Bootstrap Icons, Table | --- | --- | |
| | 6.2 Grid System, Alerts, Button, Spinner | --- | --- | |
| | 6.3 Dropdown, Cards, Collapse, Accordian | --- | --- | |
| 07 | **Bootstrap Components: Navigation & Forms** | | | 4 (15%) |
| | 7.1 **Navbar:** Dropdown, Collapsible<br>Sidebar Offcanvas components, Carousel, Modal, Scrollspy | --- | | |
| | 7.2 Pagination, **Forms:** Form Validations, Floating labels | --- | | |
| 08 | **Core JavaScript Concepts & Functions** | | | 3 (5%) |
| | 8.1 Basics of Javascript and EcmaScript(ES6),<br>Concept of Var,Let and Const Keywords, Template Literals,<br>Datatype and Variables, Array and objects,<br>**Array methods :**push(), pop(), map(), filter(), forEach(),includes(), find() | --- | --- | |
| | 8.2 **Loops**: for, while, do…while, for...of, for...in, **Condition** : if...else, switch<br>**Functions:** declarations, expressions, parameters, return values,<br>Difference between function and arrow function (ES6),<br>**Default Parameters:** Default Parameter Values,<br>Function Rest Parameter, Spread & Rest Operators | --- | --- | |
| | 8.3 **Pop up Boxes:** Alert, Confirm, Prompt | --- | --- | |
| 09 | **JavaScript DOM Manipulation & Events** | | | 3 (15%) |
| | 9.1 **DOM Manipulation:**<br>document object, getElementById, getElementsByClassName, querySelector, querySelectorAll, innerHTML, innerText, setAttribute, getAttribute, removeAttribute, style manipulation | --- | | |
| | 9.2 **Event Handling:**<br>**Events:** click, input, submit, mouse, keyboard, basic concept addEventListener,<br>**Event Object & Handling Essentials:** event.target, event.type, event.key, event.preventDefault | --- | --- | |
| 10 | **Modern JavaScript: Async, Fetch, and Forms** | | | 3 (10%) |
| | 10.1 **Form Validation:** required fields, regex, Form Processing,<br>**Storage** : Local Storage, Session Storage | **<form> & <input>(017013392–Unit-3)** | --- | |
| | 10.2 **Asynchronous JavaScript:** Introduction to Promises (.then(), .catch(), .finally()), and async/await syntax. | --- | --- | |
| | 10.3 Explain the fetch API for making HTTP requests to external APIs. | --- | --- | |

| Sr No. | Practical Title | Link to Theory Syllabus |
|---|---|---|
| 1 | Write HTML script to generate following design.<br><table><tr><td></td><td colspan="2">average</td><td rowspan="2">other category</td></tr><tr><td></td><td>height</td><td>weight</td></tr><tr><td>males</td><td>1.9</td><td>0.003</td><td>yyy</td></tr><tr><td>females</td><td>1.7</td><td>0.002</td><td>xxx</td></tr></table> | Unit-2 |
| 2 | Write HTML code to display following irregular table.<br><table><tr><td rowspan="4">A</td><td>B</td><td colspan="2">C</td><td>D</td><td colspan="2" rowspan="2">E</td></tr><tr><td>F</td><td>G</td><td rowspan="3">H</td><td>I</td></tr><tr><td rowspan="2">J</td><td>K</td><td>L</td><td>M</td><td>N</td></tr><tr><td>O</td><td>P</td><td colspan="3">Q</td></tr></table> | Unit-2 |
| 3 | Design a simple form to register for new user information and use fieldset and legend to differentiate personal details and educational details. Personal Details should contain Full Name, Email, Mobile number, Gender, Birth Date and Hobbies. and Educational Details should contain SSC Percentage, HSC Percentage, Branch(dropdown-CE,IT,CSE,CST,AIML). Add Submit and reset buttons. | Unit-3 |
| 4 | Write an HTML code to embed a video in a web document. | Unit-3 |
| 5 | Write an HTML script to draw a tricolor according to indian national flag with Ashok Chakra as a simple circle in between. | Unit-3 |
| 6 | Draw SVG Logo on a circle filled with orange color (shaded part) as shown in figure below.<br> <br> | Unit-3 |
| 7 | Apply following css using external css.<br>1)Write one paragraph using <p> tag.Color of first line of paragraph should be green and text should be in italic.<br>2)When you select any text of heading ,background color of selected text will become grey.<br>3) Create one ordered list which starts with capital roman letters. color of list should be green and font-size should be 30px. | Unit-4 |
| 8 | Create a flex container with 4 child boxes.<br>Use flex-direction: row.<br>Apply justify-content: space-between.<br>Center items vertically using align-items.<br>Give the third box order: -1 so it appears first. | Unit-4 |
| 9 | Write the following styles in separate CSS file and also show how to link this CSS file in HTML file and show use of styles.<br>i) The headings should have normal font style and font's size should be 120%.<br>ii) Define class c1 for paragraph which defines font style bold having 3cm spacing for paragraph initialization.<br>iii) Apply background color orange and apply a background image "nature.jpg" and it should be repeated horizontaly only. | Unit-4 |
| 10 | Write down HTML/CSS code that will change both the background-color and the position of the  element when the animation is 25% complete, 50% complete, 75% complete and again when the animation is 100% complete. also has a 2 seconds delay before starting the animation. an animation should be played backwards first, then forwards, the animation continue for ever. initial background color is red, then it convert to yellow, green and blue gradually. | Unit-5 |
| 11 | Create a button and perform the tasks as aked below.<br>On hover, its background color should smoothly change from blue → orange in 2 seconds.<br>Use transition-property, transition-duration, and transition-timing-function (ease-in-out).<br>Add a 1 second delay before the transition starts. | Unit-5 |
| 12 | Write down HTML/CSS code that show how to create a linear gradient (from left to right) with the color of the rainbow and some text like "Rainbow". | Unit-5 |
| 13 | Write down HTML/CSS code that changes the background-color to blue if the viewport is 250 pixels wide or wider (if the viewport is less than 250 pixels, the background-color will be pink) | Unit-5 |

| Sr No. | Practical Title | Link to Theory Syllabus |
|---|---|---|
| **14** | Design card as shown below using Bootstrap 5.<br>  <br> | Unit-6 |
| **15** | Write a script to design 3 FAQs with question and answer. Initially it shows all 3 questions only. By clicking particular question, it should display answer of that question only. If user clicks any other question, then respective answer should be opened and previous answer should disappear. | Unit-6 |
| **16** | Write a script using HTML/CSS and bootstrap to create following pagination. Make Link 1 as active & link 4 as disable.<br>  <br> | Unit-7 |
| **17** | Design following using Bootstrap5 form validation.<br> <br> | Unit-7 |
| **18** | Write an HTML using bootstrap-5 to make a set of a slide show of any three random images with buttons having a caption. It must have buttons to jump on next/previous slides also. | Unit-7 |
| **19** | Create an HTML page with a  and a button.<br>Write an ES6 function with two default parameters a and b.<br>The function should also take two arguments c and d when called.<br>On button click, the function should calculate the equation:<br>Result=(a×b)+c−d<br>Display the result inside the . | Unit-8 |
| **20** | Write a JavaScript that handles following mouse events. Add necessary elements.<br>(i) JavaScript gives the key code for the key pressed.<br>(ii) If the key pressed is “a”,”e”,”i”,”o”,”u”, the script should announce that vowel is pressed.<br>(iii)When the key is released background should change to blue. | Unit-9 |
| **21** | Write a JavaScript to handle following mouse events.<br>i.If mouse clicked on heading, content of heading changed to “Hello World” with italic effect & 50px size.<br>ii.If button named red is clicked then heading should change to red color & button named blue is clicked then heading should change to blue color. | Unit-9 |
| **22** | Create an HTML page with one heading <h1>, one paragraph <p>, a button, and an unordered list <ul>.<br>When the button is clicked, change the text of the heading.<br>Change the color of all paragraphs to red.<br>Change the background color of the first paragraph.<br>Underline all list items inside the <ul>. | Unit-9 |
| **23** | Create a calculator with two number inputs, an operation dropdown (+ - * /), and a button.<br>On button click, perform the calculation using a Promise.<br>Resolve with result if inputs are valid, reject if invalid (e.g., not numbers or division by zero).<br>Use .then() to show result, .catch() for errors, and .finally() to display "Calculation Finished". | Unit-10 |
| **24** | Design a login form using HTML & JavaScript with following validations on username and password fields.<br>1. Password length must be 6 to 12 characters<br>2. Username should not start with _, @ or number.<br>3. Both should not be blank. | Unit-10 |

| Major Components/ Equipment | |
|---|---|
| **Sr. No.** | **Component/Equipment** |
| 1 | Computer |
| 2 | Software: code: blocks, Editor, Bootstrap libraries |

| **Course Outcome** | |
|---|---|
| 1 | Understand the concepts Of WWW, HTTP Protocol and Client-Server architecture and HTML. |
| 2 | Explore the new features of CSS to define and apply CSS rules in the Web Pages for rich User Interface. |
| 3 | Determine how to build responsive web design using Bootstrap. |
| 4 | Use JavaScript to create dynamic and interactive web applications. |
| **Suggested Reference Books** | |
| 1 | HTML 5 Black Book 2Ed, By Kogent Learning Solutions Inc. |
| 2 | Javascript For Impatient Programmers By Dr. Axel Rauschmayer |
| 3 | Full Stack Javascript By Azat Mardan, Apress Publilcation |
| 4 | Spurlock Jake, Bootstrap: Responsive Web Development. O'Reilly Media, Inc. |
| 5 | Charlton Ting And R Allen Wyke, Pure Javascript By Jason D. Gilliam, Techmedia |
| **List of Open Source Software/Learning website** | |
| 1 | www.geeksforgeeks.org |
| 2 | www.tutorialspoint.com |
| 3 | www.programiz.com |
| 4 | www.codingninjas.com |

| Practical Project/Hands On Project | | |
|---|---|---|
| **Sr. No.** | **List of Practical Projects** | **Linked with Unit** |
| 1 | Design following form.<br>Registration form<br>Enter your first name: enter your first name<br>Enter your last name<br>Enter your password<br>ReEnter your password<br>Enter your email<br>Enter your mobile<br>Enter your address<br>Select your gender: male female<br>Select your hobbies: hobby1 hobby2 hobby3<br>Select your DOB: dd-mm-yyyy<br>Select your Country: Select your country<br>Upload your pic: Choose File No file chosen<br>Save My Data Reset Data | Unit 2,3 |
| 2 | Design following layout:<br>  <br> | Unit 4,5 |
| 3 | Create a webpage using Bootstrap with a button that toggles an offcanvas sidebar. The sidebar should slide in from the left or right and contain:<br>A header with a title and close button.<br>A body section with some navigation links or text content. | Unit 6,7 |
| 4 | Do as directed:<br>Write a JavaScript that handles following mouse events. Add necessary elements.<br>(i) If the mouse is over the heading, that heading should turn yellow and if the mouse goes out of the heading it should turn black.<br>(ii) If find time button is clicked show time and date information. If button named “red” is clicked, background should change to red and If button named “green” is clicked, background should change to green. | Unit 8,9 |
| 5 | Create a simple contact form with fields for username, email, and message.<br>When the form is submitted:<br>Save the entered data into localStorage.<br>Retrieve and display the stored data below the form. | Unit-10 |